

THE 12TH MAN – Equality & Fair Selection Game

Project Documentation

Team : F09

Organisation : Swecha AP

Table of Contents

1. Abstract
2. Introduction
3. Problem Statement
4. Objectives
5. Technology Stack
6. System Architecture
7. Project Modules
8. Algorithm
9. Game Flow
10. Screenshots
11. Testing
12. Results
13. Future Enhancements
14. Conclusion
15. References

1. Abstract

The 12th Man is an educational web-based simulation game that teaches fairness, equality, and merit-based team selection. Through a cricket team selection scenario, users learn the importance of making objective decisions based on performance rather than prejudice or discrimination.

2. Introduction

Background: Fair selection is essential in sports, education, and workplaces. Need for the Project: Many decisions are influenced by bias rather than merit. Importance of Fairness and Equality: Fair opportunities improve team performance and create inclusive environments.

3. Problem Statement

Discrimination and prejudice can cause deserving individuals to lose opportunities. This project demonstrates how unfair selection negatively impacts both individuals and team success.

4. Objectives

- Promote fair selection
- Teach equality
- Encourage merit-based decisions
- Increase awareness of prejudice and discrimination

5. Technology Stack

HTML5 for structure, CSS3 for styling, and JavaScript for game logic and interactivity.

6. System Architecture

Reference Architecture.png. Flow: User → User Interface → Player Data Management → Team Selection Engine → Fairness Evaluation → Match Simulation → Rewards & Analytics.

7. Project Modules

Module 1: User Interface (Start Screen, Story Screen, Trial Screen, Analysis Screen)

Module 2: Player Data Management (Player Records, Score Calculations)

Module 3: Team Selection Engine (User Selections, Validation)

Module 4: Fairness Evaluation (Ranking Players, Comparing Selections)

Module 5: Match Simulation (Fair Team Outcome, Biased Team Outcome)

Module 6: Rewards & Analytics (Badges, Statistics, Fairness Meter)

8. Algorithm

Team Selection Algorithm:

1. Load player data
2. Calculate totals
3. Sort by performance
4. Select top performers
5. Compare with user choices
6. Determine fairness
7. Display outcome

9. Game Flow

Reference Flowchart.png. Start → Story → Trials → Team Selection → Evaluation → Match Simulation → Analysis → Rewards → Summary.

10. Screenshots

Insert screenshots from the Screenshots folder: Start Screen, Trials Screen, Team Selection Screen, Analysis Screen, Reward Screen.

11. Testing

Start Game → Story displayed Continue → Next screen shown Select 11 Players → Accepted

Select less/more than 11 → Error shown Fair Selection → Team wins Biased Selection → Team loses

12. Results

Fair selections improve outcomes. Biased selections reduce team performance. Users learn equality concepts and understand the importance of objective decision-making.

13. Future Enhancements

- Multiplayer mode
- Database integration
- AI recommendations
- Mobile app version
- More sports scenarios

14. Conclusion

The project successfully teaches fairness, equality, inclusion, and merit-based selection through an engaging and interactive educational game. Users understand how fair decisions benefit both individuals and teams.

15. References

HTML Documentation